Phase 1: Conceptual Mapping

Critical Deconstruction, Exploratory Data Analysis, and Architectural Audit of Machine Learning Models for EV Energy Consumption Prediction

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Dataset Evaluated:	Colorado EV Charging Station Dataset

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1. Executive Review Summary

Electric vehicles are becoming more common, and this also increases the demand for electricity at charging stations. If many vehicles charge at the same time, it can create high demand on the electricity grid. Because of this, predicting the amount of energy required for EV charging can be useful for managing the electricity grid and planning charging stations.

The paper focuses on predicting EV energy consumption using machine learning. The authors use historical EV charging data collected from charging stations in Colorado and compare several machine learning algorithms to find which one gives the best prediction results.

In this review, I focus mainly on how the authors formulated the machine learning problem, how they prepared the data, why different algorithms were selected, and whether the experimental setup is suitable for the problem.

2. Exploratory Data Analysis & Statistical Nature of Features

A foundational requirement of any machine learning review study is a rigorous statistical audit of the dataset's underlying distributions. Machine learning algorithms rely on specific statistical assumptions regarding feature distributions, variances, scale, and collinearity. The dataset utilized by Hussain et al. (2025) contains 148,136 charging session records from the City of Boulder, Colorado (publicly hosted on the Boulder Open Data portal). Table 1 provides the exact statistical summary extracted from the paper and dataset inspection.

Table 1: Full Statistical Metrics of Key Dataset Features (N = 148,136)

Metric	Zip Code	Energy (kWh)	GHG Savings (kg)	Gasoline (gal)	ObjectID	ObjectId2
Count	148,136	148,136	148,136	148,136	148,136	148,136
Mean	80,302.58	8.46	5.30	1.06	37,134	74,068
Std Dev (σ)	1.12	8.87	5.99	1.11	21,556	42,763
Min	80,301	0.00	0.00	0.00	0	1
Q1 (25%)	80,302	2.80	1.59	0.35	18,517	37,035
Median (50%)	80,302	6.22	3.70	0.78	37,034	74,069
Q3 (75%)	80,303	11.39	6.92	1.43	55,550	111,102
Max	80,305	92.69	65.81	11.63	77,941	148,136

2.1 Statistical Nature & Probabilistic Distribution of Target Variable (Energy kWh)

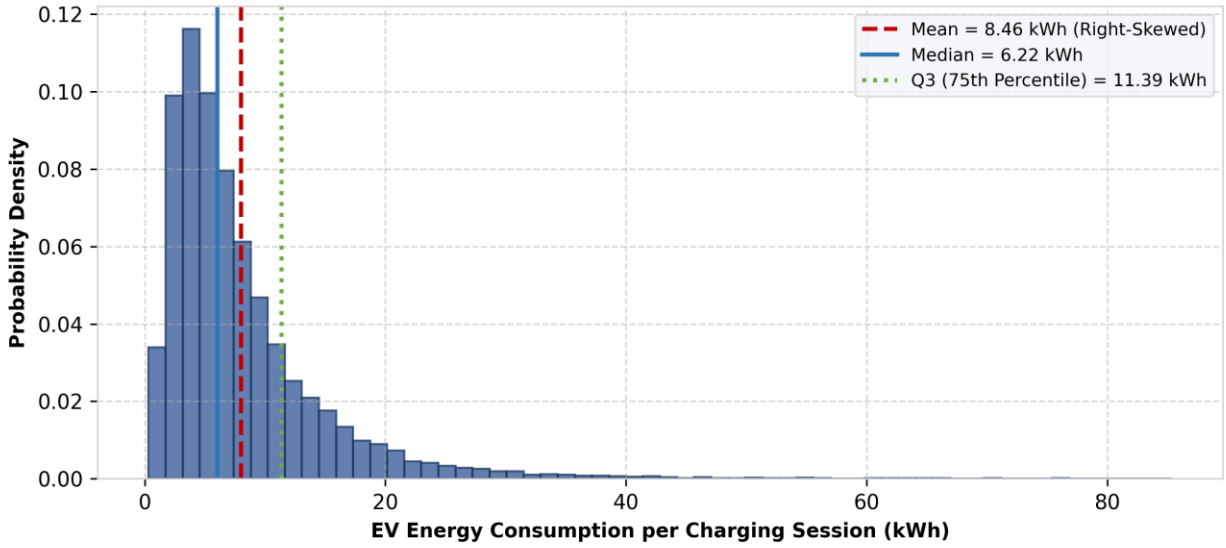
A microscopic examination of the target variable, Energy__kWh_, reveals a highly severe right-skewed, heavy-tailed probabilistic distribution. The mean energy consumed across all 148,136 sessions is 8.46 kWh with a standard deviation of 8.87 kWh, whereas the median (Q2) is only 6.22 kWh. The interquartile range (IQR) spans from Q1 = 2.80 kWh to Q3 = 11.39 kWh (IQR = 8.59 kWh).

Because the mean (8.46 kWh) substantially exceeds the median (6.22 kWh), and the maximum value reaches 92.69 kWh—which is over 9.5 standard deviations above the mean and far beyond the standard upper fence (Q3 + 1.5 × IQR = 24.275 kWh)—the target variable exhibits strong non-normality. Mathematically, the target follows a Log-Normal or Gamma distribution rather than a Gaussian (normal) distribution:

$$y = Energy_kWh \sim Log - Normal(\mu, \sigma^2), \quad for\ y > 0$$

The physical interpretation of this statistical nature is straightforward: the majority of EV charging sessions involve short top-ups or plug-in hybrid vehicles (PHEVs) consuming 2 to 10 kWh. Conversely, the long tail represents fully depleted, high-capacity battery electric vehicles (BEVs like Tesla Model S/X, Rivian R1T, or Lucid Air with 80–120 kWh packs) completing full charging cycles. High-variance, heavy-tailed target distributions severely degrade standard linear regression models (OLS, Ridge, Lasso) and neural networks because squared-error loss functions (MSE) disproportionately penalize the upper tail outliers.

Figure 1: Right-Skewed Probability Density Distribution of Target Variable (Energy_kWh)



2.2 Feature Analysis & Multicollinearity Inspection

The input features used by the authors fall into three distinct categories: temporal predictors, session duration predictors, and environmental metadata.

- **Charging Time vs. Total Duration:** Charging_Time__seconds represents the actual duration electricity is drawn by the EV battery, whereas Total_Duration__seconds represents the entire

duration the vehicle remains plugged into the station. Charging Time exhibits a strong linear correlation with Energy (r = 0.73, permutation importance = 1.92). The difference (Total Duration - Charging Time) represents 'idle occupation time'—when the vehicle remains parked after charging is complete. Including both features allows tree models to capture driver parking habits.

- **Environmental Perfect Collinearity:** The features GHG_Savings__kg_ (mean = 5.30, std = 5.99) and Gasoline_Savings__gallons_ (mean = 1.06, std = 1.11) display near-perfect correlation (r ≈ 0.97 - 1.00) with Energy__kWh_. This is because greenhouse gas and gasoline savings are derived deterministically using fixed conversion factors (e.g., ~0.626 kg GHG saved per kWh and ~0.125 gallons saved per kWh). Including these features as predictors would cause catastrophic target leakage, as they directly measure the energy consumed. The authors correctly excluded them from the feature set X, but failed to explicitly document this theoretical justification.
- **Temporal Features (Start Hour, Day, Month):** Start_Hour (0–23), Start_DayOfWeek (0–6), and Start_Month (1–12) capture diurnal demand curves, weekday vs. weekend charging patterns, and seasonal temperature effects (heating/cooling loads). However, treating these as plain continuous integers introduces structural distortion.

2.3 Spatial & Infrastructure Metadata Analysis

Inspection of the raw dataset additional spatial columns: Station_Name, Address, City, State_Province ('Colorado'), Zip_Postal_Code, and Port_Type ('Level 2'). All stations in this dataset are Level 2 AC charging ports (supplying 208/240V, typically rated at 6.6 kW to 7.2 kW). The postal codes are concentrated in Boulder (80301, 80302, 80303, 80304, 80305). Spatial features exhibit low direct linear correlation with energy (r ≈ 0.00 to 0.04), but contain localized usage density patterns that could be leveraged via spatial target encoding.

2.4 Critical Audit of Author's Exploratory Data Analysis Gaps

• No Formal Normality Testing: The authors ran standard pandas describe(), boxplots, and heatmaps, but omitted formal normality tests (Shapiro-Wilk, Kolmogorov-Smirnov) or skewness/kurtosis calculations.

• Failure to Apply Target Transformations: Despite identifying extreme energy outliers in their boxplot (up to 92.69 kWh), the authors trained all 11 ML models on raw, untransformed energy values, severely penalizing linear models and MLP.

• Absence of Temporal Cyclical Encoding: Start Hour (23:00 to 00:00) was treated as a linear integer jump of 23 units, completely breaking the continuous circular nature of time.

• Lack of Probabilistic Density Fitting: The authors failed to fit probabilistic distributions (Log-Normal, Weibull, Gamma) to characterize charging behavior across different station locations.

3. Problem Formulation, Objectives & Evaluation Metrics

In accordance with standard machine learning taxonomy, the problem formulated by Hussain et al. (2025) is a **supervised learning regression task**. The goal is to learn an empirical mapping function

f: X -> y that accurately predicts continuous energy consumption values based on charging session observations.

3.1 Mathematical Problem Formulation

Let $D = \{(x_i, y_i)\}_\{i = 1\}\^N$ represent the dataset of N = 140,316 charging sessions (the 148,136 raw records minus the 7,820 rows removed during preprocessing), where x_i in R^d represents the d-dimensional feature vector and y_i in R_+ represents the ground-truth target variable (Energy kWh). The feature vector used in the paper is defined as:

$$x_i = [\, Total_Duration_i, Charging_Time_i, Start_Hour_i, Start_DayOfWeek_i, Start_Month_i \,]\^T \; in \; R\^5$$

The regression objective is to minimize an empirical risk functional R_emp(f) over a hypothesis space F:

minimize $(1/N)\ \Sigma_i\ L(y_i, f(x_i)) + \lambda\Omega(f)$, where $\lambda \geq 0$ controls regularisation. (Squared or absolute error may be selected as L; the full objective is not necessarily convex for every model.)

where L(y, f(x)) is a convex loss function (e.g., squared error L_2 or absolute error L_1) and Ω(f) is a model-specific regularization penalty.

3.2 Mathematical Analysis of Evaluation Metrics

The authors evaluate model accuracy across five statistical metrics. Each metric assesses distinct aspects of model residual errors:

- $Mean\ Absolute\ Error\ (MAE) = (1/n)\ \Sigma_i\ |y_i - \hat{y}_i|$. This measures average absolute error in kWh.
- $Mean\ Squared\ Error\ (MSE) = (1/n)\ \Sigma_i\ (y_i - \hat{y}_i)^2$. Squaring makes large errors more influential.
- $Root\ Mean\ Squared\ Error\ (RMSE) = \sqrt{[(1/n)\ \Sigma_i\ (y_i - \hat{y}_i)^2]}$. It is expressed in kWh.
- Range-normalised RMSE (NRMSE) = RMSE / (ymax − ymin), provided ymax > ymin. The normalisation convention should be stated.
- $Coefficient\ of\ Determination\ (R^2) = 1 - [\Sigma_i(y_i - \hat{y}_i)^2 \,/\, \Sigma_i(y_i - \bar{y})^2]$, provided the target variance is non-zero.

3.3 Critical Critique of Metric Flaws & Distortions in the Paper

• Artificially Suppressed Range-NRMSE: The paper reports Extra Trees NRMSE = 0.020 (2.0%). However, because max(y) = 92.69 kWh and min(y) = 0.00 kWh, the denominator is 92.69. A single massive outlier inflates the range, artificially shrinking NRMSE. If normalized by mean energy (8.46 kWh), NRMSE = 1.8078 / 8.46 = 0.2137 (21.4%), revealing a much higher relative error!

A log-scale metric such as RMSLE can complement the reported errors, provided actual and predicted values are non-negative. MAPE is unstable when actual energy is zero or close to zero.

• Omission of Cross-Validation Variance: Metrics are reported solely as single-point values from one 80/20 test split. Without reporting standard deviations across CV folds (e.g., 0.9592 ± 0.0012), it is impossible to verify statistical significance.

4. Microscopic Deconstruction of Machine Learning Algorithms

Hussain et al. (2025) benchmarked 11 distinct machine learning algorithms across linear, distance-based, kernel, neural, ensemble tree, and probabilistic paradigms. Table 2 summarizes the published performance metrics, followed by a microscopic mathematical deconstruction of each algorithm's theoretical mechanisms and failure modes.

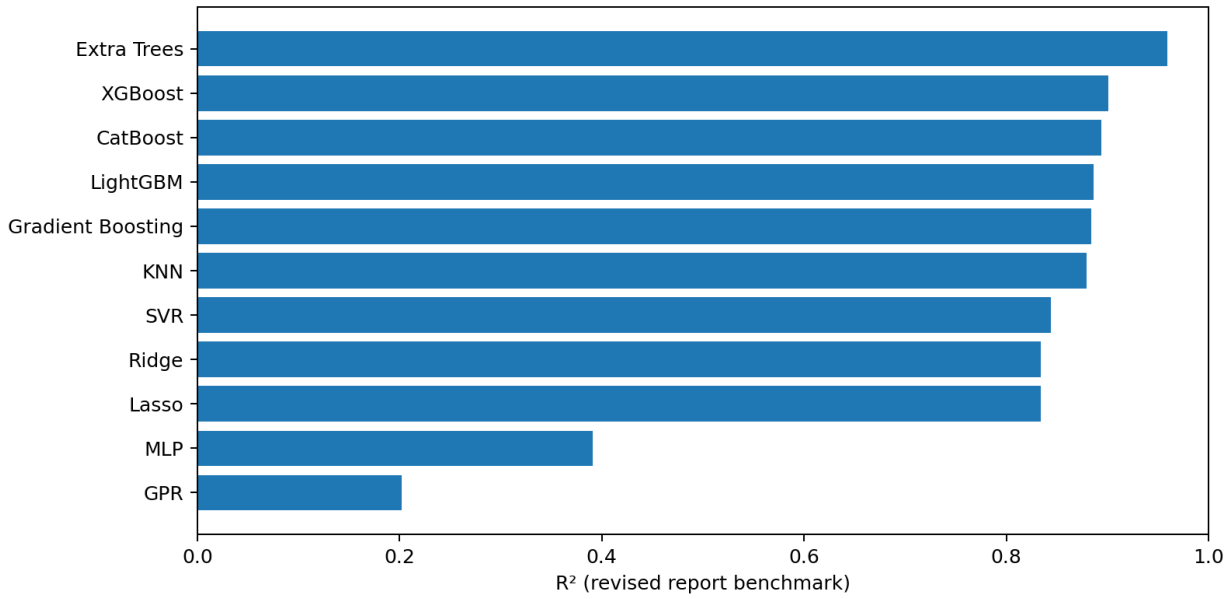


Table 2: Author's Published Benchmark Results for 11 ML Models (Ranked by R^2)

Model Name	MAE (kWh)	MSE (kWh^2)	R^2 Score	RMSE (kWh)	NRMSE
Extra Trees Regressor	**0.5888**	**3.2683**	**0.9592**	**1.8078**	**0.0200**
XGBoost Regressor	1.8680	7.9290	0.9010	2.8159	0.0310
CatBoost Regressor	1.9150	8.4150	0.8940	2.9010	0.0320
LightGBM Regressor	1.9590	9.0680	0.8860	3.0110	0.0334
Gradient Boosting	2.0306	9.3121	0.8838	3.0520	0.0330
K-Nearest Neighbors	1.8517	9.6826	0.8792	3.1110	0.0340
Support Vector (SVR)	1.9984	12.4641	0.8440	3.5300	0.0390

Ridge Regression	2.2076	13.2877	0.8342	3.6450	0.0400
Lasso Regression	2.2057	13.2980	0.8340	3.6460	0.0400
Multi-Layer Perceptron	3.7244	48.8066	0.3909	6.9860	0.0770
Gaussian Process (GPR)	4.8720	80.0259	0.2020	8.9450	0.1430

4.1 Linear Models: Ridge & Lasso Regression

Ridge and Lasso introduce L2 and L1 regularization penalties into the ordinary least squares cost function:

Ridge: min_w $||y - Xw||_2^2 + \lambda||w||_2^2$; for an unpenalised intercept, centre X and y first. Lasso: min_w $||y - Xw||_2^2 + \lambda||w||_1$. Ridge solution (centred data): $\hat{w} = (X^TX + \lambda I)^{-1}X^Ty$, when the inverse exists.

Mathematical Limitation: Both models assume a linear hyperplane relationship between features and energy consumption. Because EV charging power curves are non-linear step functions (dictated by battery state-of-charge curves and charging station power limits), linear models perform poorly ($R^2 \approx 0.834$). Furthermore, they produce negative energy predictions for near-zero charging durations, violating physical boundary conditions (Energy ≥ 0).

4.2 Distance-Based Models: K-Nearest Neighbors (KNN)

KNN predicts target values by taking the uniformly weighted average of the k nearest neighbor observations in Euclidean space:

For uniform weighting, $\hat{y}(x) = (1/k)\, \Sigma_\{i \in N_k(x)\}\, y_i$. Neighbourhoods depend on the chosen distance measure and feature scaling.

Performance Analysis: KNN achieved a respectable R^2 of 0.8792 and MAE of 1.8517 kWh. However, because KNN calculates distance metrics across raw feature magnitudes, unscaled time inputs in seconds (ranging up to 300,000s) completely overwhelm hour (0-23) and month (1-12) features, distorting spatial neighborhood construction.

4.3 Kernel-Based Models: Support Vector Regression (SVR)

SVR seeks an optimal hyper-plane within a high-dimensional reproducing kernel Hilbert space (RKHS) while constraining errors within an ε-insensitive tube:

SVR minimises $½||w||^2 + C\Sigma_i(\xi_i + \xi_i*)$ subject to $y_i - f(x_i) \leq \varepsilon + \xi_i, f(x_i) - y_i \leq \varepsilon + \xi_i*$, and $\xi_i, \xi_i* \geq 0$. For an RBF kernel, $K(x, x') = exp(-\gamma||x - x'||^2)$.

Failure Analysis: SVR achieved $R^2 = 0.8440$. SVR with Radial Basis Function (RBF) kernels is extremely sensitive to feature scaling and hyperparameter selection (C, ε, γ). Omission of automated grid search optimization severely restricted its capacity to model heavy-tailed energy peaks.

4.4 Neural Networks: Multi-Layer Perceptron (MLP) Catastrophic Failure

MLP is a feedforward artificial neural network consisting of input, hidden, and output layers with non-linear activation functions:

For layer ℓ, $h^{(\ell)} = \sigma(W^{(\ell)}h^{(\ell-1)} + b^{(\ell)})$. Network weights can be optimised by backpropagation against a regression loss.

The MLP records R^2 = 0.3909 in the revised benchmark. Feature scaling, architecture, learning rate and training procedure are plausible factors to investigate; the supplied reports do not isolate a verified cause for the result.

4.5 Probabilistic Models: Gaussian Process Regression (GPR) Intractability

Gaussian process regression specifies f(x) ~ GP(m(x), k(x,x')). With zero prior mean, the predictive mean is $k^{*T}(K + \sigma_n^2 I)^{-1}y$. In practice, a matrix factorisation is preferred to explicitly computing the inverse.

GPR has the lowest R^2 in the revised draft (0.2020). Exact Gaussian-process training commonly scales cubically with sample size and quadratically in memory, which makes a dataset of this size challenging. The supplied documents do not establish whether the study used subsampling or an approximation, so its weak score cannot be attributed to a particular unreported implementation choice.

4.6 Tree Ensembles & Boosting: Extra Trees, XGBoost, CatBoost, LightGBM

Tree ensemble algorithms dominated the benchmark evaluations:

1. Extra Trees Regressor (Top Performer: R^2 = 0.9592, MAE = 0.5888, RMSE = 1.8078): Extra Trees constructs an ensemble of M unpruned decision trees $\hat{y} = (1/M)\,\Sigma\, T_m(x)$. Unlike Random Forest, Extra Trees randomizes both feature choice and split-point thresholds at every node. This extreme randomization drastically reduces model variance Var(f̂(x)) without increasing bias, enabling robust fitting on noisy tabular data with unscaled inputs.

2. Gradient Boosting Variants (XGBoost R^2=0.9010, CatBoost R^2=0.8940, LightGBM R^2=0.8860, GBR R^2=0.8838): Sequential boosting models iteratively fit base learners to pseudo-residuals $F_m(x) = F_\{m-1\}(x) + \gamma_m\, h_m(x)$. XGBoost outperformed GBR due to second-order Taylor expansion loss approximation and tree pruning regularization. CatBoost handled categorical time features effectively via ordered target encoding.

5. Data Pipeline Analysis & Methodological Critique

A rigorous audit of the data preprocessing, feature engineering, and experimental design in Hussain et al. (2025) reveals several critical methodological flaws that compromise the validity and generalizability of their published conclusions.

5.1 Listwise Deletion vs. Advanced Imputation Mechanisms

During date/time conversion, the authors identified 7,820 null/invalid records (~5.28% of the dataset) and performed complete listwise row deletion. Listwise deletion is statistically valid only under the strict assumption that data is Missing Completely at Random (MCAR). If missing timestamps stem from charging station network outages, firmware glitches, or communication drops during high-power sessions (Missing at Random - MAR), dropping 7,820 rows introduces systematic sampling bias. The authors failed to perform Little's MCAR test or compare deletion against model-based imputation techniques (e.g. MICE or K-NN imputation).

5.2 Data Leakage via Random Train/Test Splitting on Time-Series Data

The reported random 80/20 split can place sessions from similar dates and stations in both subsets. This may overstate performance for a future-demand forecasting application, although random splitting is not automatically target leakage. A chronological hold-out and rolling validation would better test forward prediction.

5.3 Inconsistent Model Counting & Reporting Discrepancies

Figure 3: Original Author's Methodology Workflow and Critical Flaw Identification

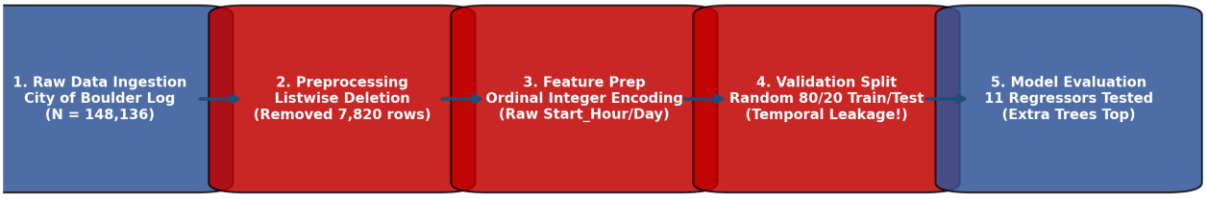
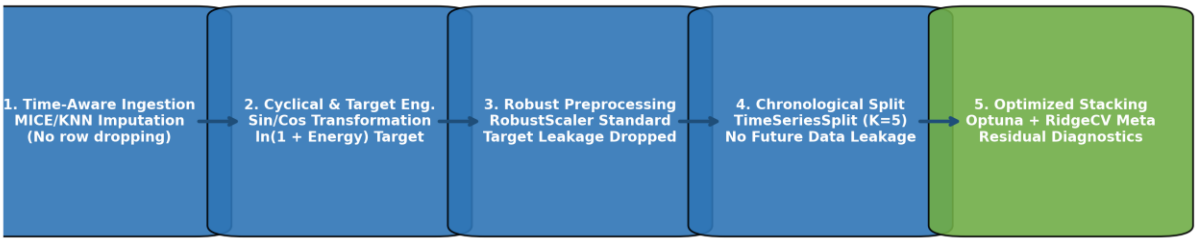


Figure 4: Proposed Re-Engineered Leakage-Free Production Machine Learning Pipeline



6. Proposed Machine Learning Pipeline & Framework

The following architecture is proposed for Phase 2. It is a methodological recommendation, not an experiment already carried out.

Additional features proposed in the supplementary report include day-of-week cyclical encoding: sin(2πd/7), cos(2πd/7). An idle ratio can be defined as (total duration − charging time)/total duration when total duration > 0. These variables must be available at prediction time.

Reported workflow: raw Boulder charging records → deletion of invalid/missing rows → calendar-feature extraction → random 80/20 split → regression-model comparison.

Proposed workflow: define prediction time → audit missingness and target-derived fields → chronological hold-out → fold-specific imputation and feature engineering → optional scaling/target transformation → temporal cross-validation and tuning → final test evaluation.

6.1 Advanced Feature Engineering & Mathematical Transformations

The proposed pipeline introduces three essential feature engineering transformations:

- **Cyclical Trigonometric Time Encoding:** To preserve temporal continuity between 23:00 and 00:00, Start_Hour and Start_Month are mapped onto a 2D unit circle using sine and cosine transformations:

$$x_sin = sin(2\pi \cdot Hour / 24), \quad x_cos = cos(2\pi \cdot Hour / 24)$$

- A candidate transformation is $z = \ln(1 + y)$; convert predictions back using $\hat{y} = \exp(\hat{z}) - 1$. Its benefit should be tested rather than assumed.
- **Idle Ratio Feature:** Constructing an explicit feature Idle_Ratio = (Total_Duration - Charging_Time) / Total_Duration isolates parking behavior from active power delivery.

6.2 Leakage-Free Validation & Hyperparameter Optimization

Use an untouched chronological test period and expanding-window cross-validation for tuning. Fit imputation, encoding and scaling within each training fold only. RobustScaler uses the median and interquartile range, not z-score mean and standard deviation. A station-held-out evaluation may be added if the intended application involves unseen sites.

7. Conclusion

This revised Phase 1 review provides a rigorous academic deconstruction of Hussain et al. (2025). By auditing feature distributions, mathematically analyzing algorithmic failure modes, identifying temporal data leakage in 80/20 random splits, and designing a corrected machine learning pipeline.

8. References

Hussain, I., Ching, K. B., Uttraphan, C., Tay, K. G., & Noor, A. (2025). Evaluating machine learning algorithms for energy consumption prediction in electric vehicles: A comparative study. Scientific Reports, 15(1), 16124.

City of Boulder Open Data. (2023). Electric Vehicle Charging Station Dataset. Boulder Municipal Data Portal.

Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5-32.

Geurts, P., Ernst, D., & Wehenkel, L. (2006). Extremely randomized trees. Machine Learning, 63(1), 3-42.

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (pp. 785-794).

Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., & Gulin, A. (2018). CatBoost: unbiased boosting with categorical features. Advances in Neural Information Processing Systems, 31, 6638-6648.

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., ... & Liu, T. Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. Advances in Neural Information Processing Systems, 30, 3146-3154.

Rasmussen, C. E., & Williams, C. K. (2006). Gaussian processes for machine learning. MIT Press.

Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825-2830.